

STATISTICAL BUSINESS ANALYSIS REPORT

Week 7 · Introduction to Statistics for Data Science

EXECUTIVE SUMMARY

This report presents a comprehensive statistical analysis of customer churn and sales data. The analysis includes descriptive statistics, distribution analysis, correlation analysis, hypothesis testing, confidence intervals, and regression analysis.

KEY METRIC	VALUE
Churn Rate	10.60%
Average Monthly Charges	\$113.64 ± \$4.55 (95% CI)
Average Tenure	36.53 ± 1.82 months (95% CI)
Average Total Sales	\$123,650.48 ± \$19,874.13 (95% CI)

1. DATA OVERVIEW

DATASET	RECORDS	KEY VARIABLES
Customer Churn Data (customer_churn.csv)	500 customers	CustomerID, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen, Churn
Sales Data (sales_data.csv)	100 transactions	Date, Product, Quantity, Price, Customer_ID, Region, Total_Sales

2. DESCRIPTIVE STATISTICS

Customer Churn Data

STATISTIC	TENURE	MONTHLY CHARGES	TOTAL CHARGES
Mean	36.53	\$113.64	\$4,237.88
Median	37.00	\$115.00	\$4,182.50

Mode	70	\$199	\$7,693
Std Deviation	20.67	\$51.80	\$2,260.62
Minimum	1	\$20	\$159
Maximum	71	\$199	\$7,993
Skewness	-0.12	0.04	0.11
Kurtosis	-1.20	-1.21	-1.18

Sales Data

STATISTIC	QUANTITY	PRICE	TOTAL SALES
Mean	4.78	\$25,808.51	\$123,650.48
Median	5.00	\$24,192.00	\$97,955.50
Mode	9	\$43,950	\$7,148
Std Deviation	2.59	\$13,917.63	\$100,161.09
Minimum	1	\$1,308	\$6,720
Maximum	9	\$49,650	\$773,750
Skewness	-0.10	0.26	2.68
Kurtosis	-1.24	-0.95	7.68

3. DATA DISTRIBUTION ANALYSIS

Normality Tests (Shapiro-Wilk)

VARIABLE	W-STATISTIC	P-VALUE	NORMAL ($\alpha=0.05$)
Tenure	0.9563	0.0000	No
MonthlyCharges	0.9549	0.0000	No
TotalCharges	0.9701	0.0001	No
Quantity	0.9565	0.0033	No
Price	0.9788	0.0963	Yes
Total_Sales	0.7659	0.0000	No

Interpretation: Most variables do not follow a normal distribution, which is common in business data. This affects the choice of statistical tests used in hypothesis testing.

4. CORRELATION ANALYSIS

Pearson Correlations — Customer Churn Data

	TENURE	MONTHLY CHARGES	TOTAL CHARGES	CHURN
Tenure	1.0000	-0.0104	-0.0057	-0.5092
MonthlyCharges	-0.0104	1.0000	-0.0423	0.1074
TotalCharges	-0.0057	-0.0423	1.0000	0.0342
Churn	-0.5092	0.1074	0.0342	1.0000

Pearson Correlations — Sales Data

	QUANTITY	PRICE	TOTAL SALES
Quantity	1.0000	0.0317	0.6882
Price	0.0317	1.0000	0.6464
Total_Sales	0.6882	0.6464	1.0000

Key Correlation Findings

- Tenure & Churn:** $r = -0.5092$ (STRONG NEGATIVE) — Customers with longer tenure are significantly less likely to churn ($p < 0.001$).
- Monthly Charges & Churn:** $r = 0.1074$ (WEAK POSITIVE) — Higher monthly charges slightly increase churn likelihood ($p = 0.016$).
- Quantity & Total Sales:** $r = 0.6882$ (STRONG POSITIVE) — Higher quantities sold lead to substantially higher total sales.

- **Price & Total Sales: $r = 0.6464$ (MODERATE-STRONG POSITIVE)** — Higher-priced items generate meaningfully higher sales revenue.

5. HYPOTHESIS TESTING

TEST 1: Monthly Charges by Churn Status		Independent Two-Sample T-Test
H ₀ : No difference in monthly charges between churned and non-churned		H ₁ : Significant difference exists
Statistic: 2.4102		P-value: 0.016303
Significant ($\alpha=0.05$): YES ✓		<i>Customers who churn have significantly higher monthly charges than those who don't.</i>
TEST 2: Tenure by Churn Status		Independent Two-Sample T-Test
H ₀ : No difference in tenure between churned and non-churned		H ₁ : Significant difference exists
Statistic: -13.2034		P-value: < 0.000001
Significant ($\alpha=0.05$): YES ✓		<i>Customers who churn have significantly shorter tenure. Critical finding for retention.</i>
TEST 3: Sales by Product Category		One-Way ANOVA
H ₀ : All product categories have equal mean sales		H ₁ : At least one category differs
Statistic: F = 2.0508		P-value: 0.093428
Significant ($\alpha=0.05$): NO ✗		<i>No significant difference in sales across product categories — balanced revenue distribution.</i>
TEST 4: Monthly Charges vs Hypothesized Mean (\$100)		One-Sample T-Test
H ₀ : Population mean = \$100		H ₁ : Population mean $\neq$ \$100
Statistic: 5.8863		P-value: < 0.000001
Significant ($\alpha=0.05$): YES ✓		<i>Average monthly charges (\$113.64) is significantly higher than the hypothesized \$100.</i>
TEST 5: East vs West Regional Sales		Independent Two-Sample T-Test
H ₀ : No difference between East and West regions		H ₁ : Significant difference exists
Statistic: 2.0202		P-value: 0.049619
Significant ($\alpha=0.05$): YES ✓		<i>Significant difference in sales between East and West; East shows higher sales.</i>

6. CONFIDENCE INTERVALS (95%)

VARIABLE	MEAN	MARGIN OF ERROR	95% CI
Tenure (months)	36.53	± 1.82	[34.72, 38.35]
MonthlyCharges (\$)	\$113.64	$\pm \$4.55$	[\$109.08, \$118.19]
TotalCharges (\$)	\$4,237.88	$\pm \$198.63$	[\$4,039.25, \$4,436.51]
Quantity	4.78	± 0.51	[4.27, 5.29]
Price (\$)	\$25,808.51	$\pm \$2,761.56$	[\$23,046.95, \$28,570.07]
Total_Sales (\$)	\$123,650.48	$\pm \$19,874.13$	[\$103,776.35, \$143,524.61]

Interpretation: We are 95% confident that the true population means fall within these intervals. For example, the true average monthly charges lie between \$109.08 and \$118.19.

7. REGRESSION ANALYSIS

MODEL	EQUATION	R ²	INTERPRETATION
Tenure → TotalCharges	TotalCharges = 4260.57 + (−0.62) × Tenure	0.0000	Tenure has virtually no predictive power for TotalCharges.
MonthlyCharges → TotalCharges	TotalCharges = 4447.56 + (−1.85) × MonthlyCharges	0.0018	MonthlyCharges has minimal impact on TotalCharges.
Price → Total_Sales	Total_Sales = 3640.54 + 4.65 × Price	0.4175	Price explains ~42% of Total_Sales variance. Each \$1 increase in Price leads to a \$4.65 increase in Total_Sales.
Quantity → Total_Sales	Total_Sales = −3638.74 + 26629.54 × Quantity	0.4735	Quantity explains ~47% of Total_Sales variance. Each additional unit sold increases Total_Sales by \$26,629.54.

8. BUSINESS INSIGHTS & RECOMMENDATIONS

CHURN RISK FACTORS	- Customers with shorter tenure are significantly more likely to churn - Higher monthly charges are associated with higher churn risk - Target customers with tenure < 12 months for retention programs
SALES PATTERNS	- No significant difference in sales across product categories - Both quantity and price are moderate-to-strong predictors of total sales - East region shows higher sales than West region
CUSTOMER VALUE	- Average customer tenure: 36.53 months - Average monthly charges: \$113.64 - Average total charges: \$4,237.88

Strategic Recommendations

AREA	RECOMMENDATION
RETENTION STRATEGY	Implement loyalty programs for customers with tenure < 12 months; offer discounts to increase early-stage retention.
PRICING OPTIMIZATION	Review pricing strategy — customers paying > \$130/month show higher churn; consider tiered pricing for higher-value segments.
PRODUCT STRATEGY	Maintain balanced product portfolio (no significant sales differences); focus on increasing quantity per transaction.
REGIONAL FOCUS	Investigate why East region outperforms West and apply successful strategies to West region.
CUSTOMER LIFETIME VALUE	Focus on increasing tenure to reduce churn; long-term customers generate more stable revenue.

9. METHODOLOGY

CATEGORY	METHODS / TOOLS
Statistical Methods	Descriptive statistics (mean, median, mode, std dev, variance, range, IQR, skewness, kurtosis); distribution analysis
Tools	Python (pandas, numpy, scipy, matplotlib, seaborn, scikit-learn); Jupyter Notebook

Significance Level	$\alpha = 0.05$
Datasets	customer_churn.csv (500 records) · sales_data.csv (100 records)

10. CONCLUSION

This comprehensive statistical analysis provides valuable insights into customer churn patterns and sales performance. The key findings are summarised below:

- Tenure is the strongest predictor of churn — customers with shorter tenure are significantly more likely to churn.
- Higher monthly charges are associated with higher churn risk.
- Sales performance shows regional variations with the East region outperforming West.
- Both quantity and price are significant predictors of total sales.

These insights can guide data-driven business decisions for customer retention and sales optimisation.

Report Generated: 2024 · Analysis: Week 7 Statistical Analysis Project · Datasets: customer_churn.csv (500 records), sales_data.csv (100 records)